

SQL Joins - Dataset and Questions

```
-- Create Database
CREATE DATABASE CollegeDB;
USE CollegeDB;

-- Students Table
CREATE TABLE Students (
    student_id INT PRIMARY KEY,
    name VARCHAR(50),
    age INT,
    department_id INT
);

-- Courses Table
CREATE TABLE Courses (
    course_id INT PRIMARY KEY,
    course_name VARCHAR(100),
    department_id INT,
    instructor_id INT
);

-- Enrollments Table (Many-to-Many)
CREATE TABLE Enrollments (
    enrollment_id INT PRIMARY KEY,
    student_id INT,
    course_id INT,
    grade VARCHAR(2)
);

-- Instructors Table
CREATE TABLE Instructors (
    instructor_id INT PRIMARY KEY,
    name VARCHAR(50),
    department_id INT
);

-- Departments Table
CREATE TABLE Departments (
    department_id INT PRIMARY KEY,
    department_name VARCHAR(100)
);
```

-- Insert Sample Data

INSERT INTO Departments VALUES

(1, 'Computer Science'),
(2, 'Electrical Engineering'),
(3, 'Mathematics');

INSERT INTO Students VALUES

(101, 'Alice', 20, 1),
(102, 'Bob', 21, 2),
(103, 'Charlie', 22, 1),
(104, 'Diana', 23, 3);

INSERT INTO Instructors VALUES

(201, 'Dr. Smith', 1),
(202, 'Dr. Johnson', 2),
(203, 'Dr. Williams', 3);

INSERT INTO Courses VALUES

(301, 'Database Systems', 1, 201),
(302, 'Circuits', 2, 202),
(303, 'Linear Algebra', 3, 203),
(304, 'Data Structures', 1, 201);

INSERT INTO Enrollments VALUES

(401, 101, 301, 'A'),
(402, 102, 302, 'B'),
(403, 103, 301, 'B'),
(404, 104, 303, 'A'),
(405, 101, 304, 'A'),
(406, 103, 304, 'C');

1. List all courses with their instructor names.
2. Show student names along with the course names they are enrolled in.
3. Show course names and corresponding department names.
4. List all students with their grades and course names.
5. Show instructors and the courses they teach.
6. Show the name of students who are enrolled in "Data Structures".
7. List all students and their instructors for each course.
8. Show all students from the "Computer Science" department.
9. Show all courses taught by instructors from the "Mathematics" department.
10. List students who have received an 'A' grade.
11. Show students enrolled in courses taught by "Dr. Smith".
12. Display course names along with the number of students enrolled.
13. List departments and the number of students in each.
14. List courses that have more than one student enrolled.
15. Show names of students not enrolled in any course.
16. Show instructors who are not teaching any course.
17. Show all students and all courses, even if the student is not enrolled (FULL OUTER JOIN emulation).
18. List all students enrolled in courses from departments other than their own.
19. List all students and the total number of courses they are enrolled in.